



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 6 • STANDARD 6.AT.7

Simplify Algebraic Expressions

Lesson 6-7 • Teacher Copy

Name: _____ **Date:** _____ **Class:** _____



TODAY'S OBJECTIVES

Content Objective: I can simplify algebraic expressions by combining like terms.

Language Objective: I can explain my steps using the words like terms, combine, simplify, and coefficient.

See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Like Terms

Combine

Simplify

Coefficient

Term

Distributive property



Tap any word to see what it means and a picture.

1 Terms with the same variable part, like $3y$ and $8y$, are .

2 To add or subtract like terms into a single term is to them.

3 To write an expression in its shortest equal form is to it.

4 The number multiplied by a variable is the .

5 A number or variable separated by $+$ or $-$ signs is a .

6 Multiplying each part of a sum by a number uses the property.

Reflect — Exit Ticket

Simplify $6x + 3 + 2x + 5$.

- A. $8x + 8$
- B. $8x^2 + 8$
- C. $62x + 35$
- D. $16x$

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Like Terms
2. **Notes 2:** Combine
3. **Notes 3:** Simplify
4. **Notes 4:** Coefficient
5. **Notes 5:** Term
6. **Notes 6:** Distributive property
7. **Watch for this mistake:** *A common mistake in Simplify Algebraic Expressions is skipping the key idea: "Only combine terms that have the same variable part; numbers without a variable combine with each other." — always check your work against this rule before you submit.*
8. **Try It 1:** A. $4y$ — *Subtract the coefficients of the like terms: $7 - 3 = 4$. Keep the variable y , giving $4y$.*
9. **Try It 2:** A. $9m + 4$ — $6m + 3m = 9m$. The constant 4 has no like term, so it stays. Result: $9m + 4$.
10. **Exit Ticket:** A. $8x + 8$ — $6x + 2x = 8x$ and $3 + 5 = 8$, so the simplified expression is $8x + 8$.